# My Mobile, My Guide

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**Think your PDAs and cell phones are essentials? They're about to become indispensible**

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You're bored. Quite understandable—it happens to everyone. Worse still, you're in a city that you don't call home. What do you do? If you've got GPRS, you could whip out your mobile phone and consult the almighty Google—"leisure activities [your current location]", you'd type, and browse through the results. It would do well to be sitting down, because these things take time—picking a worthy site among all that rubbish can be tedious. Still, the information exists, which is what matters in the end.

But what if your phone would do that for you, without your intervention? What if it detected that you're not in your home city and would eventually feel the need to see the sights? What if it told you that you'd just crossed the city's best pub, and almost missed out on the experience? Call it Context Aware Computing or Ubiquitous Computing or whatever name that catches your fancy, the fact remains that with some connectivity and a lot of information, such scenarios won't remain the product of an over-active imagination.

## Context?

In geek speak, *context awareness* is the ability of a system (a nice, generic term for a piece of hardware that runs software) to be able to recognise its environment—location, temperature, altitude, that sort of thing—and act accordingly. A GPS device is one of the simplest context-aware computer—it's aware of your position, and based on that, is able to tell you what time the sun will rise or set where you are.

Context-aware Computing is just another piece in a much larger (and more Utopian) jigsaw puzzle called Ubiquitous Computing. The term was coined back in 1988 by Mark Weiser, who was the chief scientist at Xerox's PARC (Palo Alto Research Centre). The idea is that all technology around us—down to the humble light-bulb—will connect to, and interact with each other, and adapt to suit your needs. Biometric chips in your clothes will tell your air conditioner that you're too cold, and the AC will turn up the temperature to make you feel better, for example. You could cook up a million scenarios like this, and they'll all be plausible in a world with ubiquitous computing, and therein lies the catch—it'll likely be many years before this could happen.





Context-aware computing is the realist's answer to ubiquitous computing—it doesn't approach the dream, but is far better than current reality...

## Starting Small

Think Hitchhikers' Guide to The Galaxy—not the book, but the book *in* the book (wrap your heads around that one)—the single source for all the information you need as you travel the galaxy—from how to deal with an irate Vogon to the recipe for a Pan Galactic Gargle Blaster. We've got something like it already in development—it's called the Wikipedia.

Now consider that right next to many places of interest on Google Earth, you'll see a little Wikipedia icon, which shows you the Wikipedia entry on that place when you click on it. Not so on Google Maps, and WikiMapia would fit the bill if it weren't full of landmarks created by people saying "Look, I live here!"

Finally, consider that GPS-enabled cell phones are readily available in the market, and are getting cheaper with every new model. Throw in the high-speed mobile connectivity we yearn so much for—you see where this is going, don't you? The GPS in your phone will tell Google Earth (or Google Maps) what position to centre on, and you can then read the Wikipedia entries on important places in your area (not as helpful if you're not out sightseeing, but anything that gets rid of annoying guides prattling on can only be good). Bring the Wikipedia overlay into Google Maps, and this will happen *now*—on the iPhone and the N95, for instance, in countries where respectable mobile Internet speeds are not too much to ask for.

It's started.

## Being Polite

Remember when the cell phone first entered our lives? How we'd cringe every time a phone blared out an ugly ring or a cheap imitation of a song? Cellular etiquette is as important, and as shamelessly ignored as it was back then, but there is hope in sight. Researchers at Intel are working on software for your cell phone that'll make it more aware of your surroundings, and even decide whether to interrupt with a phone call or take a voicemail message instead.

The software will analyse your speech—not the words themselves, but the tone, pitch, volume, and the time for which you go on—and the speech of the people around you, and determine what sort of situation you're in. For example, a conversation between you and your friend will have a casual tone, and since you're friends who don't mind interrupting each other, the conversation will have a lot of overlapping speech. If you've told the software that it's all right to interrupt casual conversations, you'll feel that familiar buzzing when you get a call. If, on the other hand, you're in a meeting, the phone will detect the formal, subdued tones, and tell the caller to leave a voicemail message instead. At the very least, it could automatically switch to Vibrate. The software will also be able to detect your mood based on how you speak, and act accordingly—not interrupt you when you're ticked, for instance.

At this stage, social interactions are still being analysed to help the software make decisions better—it isn't particularly accurate with mood detection (and we doubt it'll go beyond a point), and hasn't been tested on a cell phone. Indeed, even the most powerful phone today can't stand up to the punishment of getting all that data from a

Intel used this unobtrusive sensor to collect data about the wearer's social interactions in their trials